Mechanics

- Everything in our universe is in a state of motion. Our solar system moves through space in the Milky Way Galaxy. Earth revolves around the Sun while rotating about its own axis. People, animals, air, and countless other objects move about on Earth's surface. The elementary particles that make up all matter, too, are constantly in motion.
- Motion: is a change in the position of an object with time
- Mechanics is a branch of Physics that deals with motion.
- There are basically two branches of mechanics:

(i) **Statics**: where we deal with objects at rest or in equilibrium under the action of balanced forces and,

(ii) **Kinetics**: which deals with actual motion.

Motion

- Kinetics can be further divided into two branches **(i) Kinematics & (ii)Dynamics.**

(i) kinematics, Describe various motions without discussing their cause. Various parameters discussed in kinematics are distance, displacement, speed, velocity and acceleration.

(ii) Dynamics: Describe the motion along with its cause, which is force and/or torque, momentum, energy, power, etc. in addition to those in kinematics.

- The motion of objects can be divided in **three** categories:

1. **Motion in One Dimension (Rectilinear Motion)**

2. **Motion in Two Dimension**

3. **Motion in Three Dimension**

Rectilinear Motion

kinematics

- Motion along a straight line, the force acting on the object and the velocity of the object both are along one and the same line. The distances are measured along the line only and we can indicate distances along the +ve and –ve axes as being positive and negative, respectively.

- To describe motion along a straight line (one dimension), we use:

Quantity	Symbol	Definition	Units
Position	x̄	Location of a particle	m
Displacement	Δx̄	Change in position	m
Velocity	V	Rate of change of position (speed + direction)	m/s
Acceleration	A	Rate of change of velocity	m/s^2

- **A. Position, Displacement, and Distance**

i. Position ($\bar{x}$): The location of an object at a particular time, measured with respect to a chosen origin (x=0) on a coordinate axis (e.g., an x-axis). It is a vector quantity because direction from the origin matters.

Example: x = +3 m means the object is 3 meters to the right of the origin.

ii. Displacement ($\Delta\bar{x}$): The change in position of an object. It is a vector quantity defined as:

$\Delta\bar{x} = \bar{s} = \bar{x}_2 - \bar{x}_1$ ($\bar{x}_2$ is the final position and $\bar{x}_1$ is the initial position.)

It Depends only on the initial and final positions, not on the path taken. A negative displacement means the object moved in the negative direction.

iii. Distance (d): The total length of the actual path traveled. It is a scalar quantity (always positive).

If an object travels along the *x*-axis from (*x* = 2m) to (*x* = 5m) then the distance travelled is (3m). In this case the displacement is also (3m) and its direction is along the +ve x-axis

- **B. Velocity and Speed:** Velocity and speed describe how position changes with time.
- **Average Velocity (v_{avg}):** The displacement divided by the time interval during which that displacement occurred. It is a vector quantities.

$$v_{avg} = \frac{\overrightarrow{x_2} - \overrightarrow{x_1}}{(t_2 - t_1)} = \frac{\Delta\vec{x}}{\Delta t}$$

The sign of the average velocity indicates the direction of motion.

ii. Average Speed (s_{avg}): The total distance traveled divided by the total time interval. It is a scalar quantity and has the same dimensions as that of velocity. i.e.$[L^1\, M^0\, T^{-1}]$.

$$\boldsymbol{s_{avg}} = \frac{\text{total distance}}{\Delta t}$$

e.g.: Average velocity cares about the net change; average speed cares about the total path length. If you run a lap and return to your starting point, your average velocity is zero, but your average speed is not.

iii. Instantaneous Velocity (v): This is the velocity of an object at a single instant in time.

$$\vec{v} = \lim_{\Delta t \to 0} \left(\frac{\Delta \bar{x}}{\Delta t}\right) = \frac{d\bar{x}}{dt}$$

Mathematically, It is defined as the slope of the tangent line to the position Vs time graph at that point. $\vec{v} = \frac{d\bar{x}}{dt}$

(the derivative of position with respect to time)

iv. Instantaneous Speed (s): The magnitude (absolute value) of the instantaneous velocity. It is a scalar and is always non-negative.

$$\mathrm{v} = |\vec{v}|$$

v. Relative Velocity: Relative velocity of object A with respect to object B as the difference between their velocities, i.e., $V_{AB} = V_A - V_B$

Similarly, the velocity of B with respect to A is given by: $V_{BA} = V_B - V_A$

We assume that at time *t* = 0, A and B were at the same point *x* = 0. As they are travelling with different velocities, the distance between them

will go on increasing with time in direct proportion to the difference in their velocities, v i.e., the relative velocity between them.

- **C. Acceleration:** Describes how velocity itself changes with time. It is a vector quantity and its dimensions are $[L^1M^0T^{-2}]$

i. Average Acceleration (a_{avg}): The change in velocity divided by the time interval over which that change occurred.

$$\vec{a}_{avg} = \frac{\Delta\vec{v}}{\Delta t} = \left(\frac{\vec{v_2}-\vec{v_1}}{t_2-t_1}\right)$$

Important: Acceleration is often misunderstood. It does not always mean "speeding up." ($\vec{a}$ and $\vec{v}$ same sign: Object is speeding up. $\vec{a}$ and $\vec{v}$ opposite signs: Object is slowing down (decelerating)).

A constant, non-zero acceleration means the velocity is changing at a steady rate.

ii. Instantaneous Acceleration (a): The acceleration at a specific instant. Is the limiting value of the average acceleration when the time interval goes to zero. It is given by

$$\vec{a} = \lim_{\Delta t \to 0}\left(\frac{\Delta\vec{v}}{\Delta t}\right) = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{x}}{\mathrm{d}t^2}$$

Mathematically, it is the derivative of velocity with respect to time and the second derivative of position.

On a velocity-time (v-t) graph, it is the slope of the tangent line.

- You can feel acceleration (e.g., in a starting or stopping elevator), but you cannot feel constant velocity.
- Notes
  - Average velocity = (total displacement)/(time).

- Instantaneous velocity = slope of position–time graph.
- Uniform motion → velocity constant, acceleration = 0.
- Non-uniform motion → velocity changes, acceleration ≠ 0.
- **Life-science example:**

A white blood cell crawling through tissue moves only a few micrometers per second, yet its motion obeys the same mathematical definitions as a falling apple.(When we Calculate speed and acceleration).

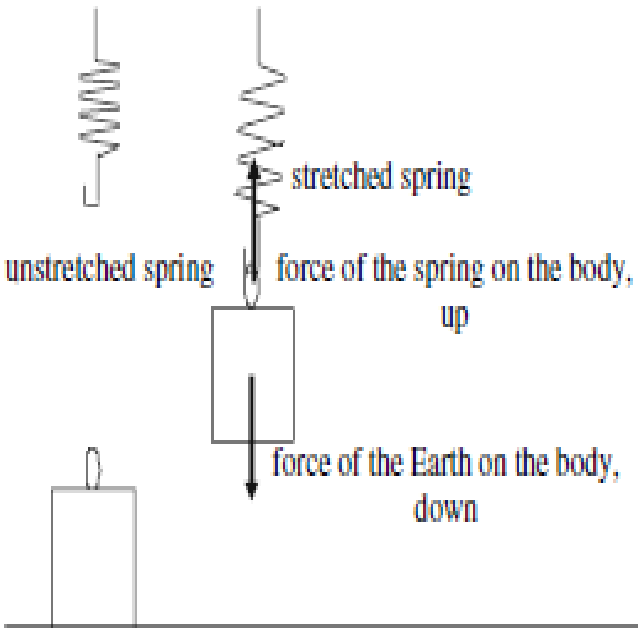
- **Force in One Dimension:**
- Forces :A force is that which causes an acceleration .

Formulated by Sir Isaac Newton (1642 - 1727).

- **A force:** is a push or a pull, representing an interaction

between two objects. It is measured in newtons (N).

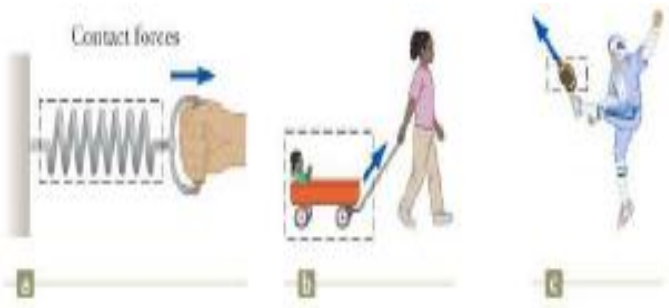
It can be measured using a calibrated spring. The stretch of the spring is proportional to the force it exerts (e.g., a grocery scale).



Classes of Forces:

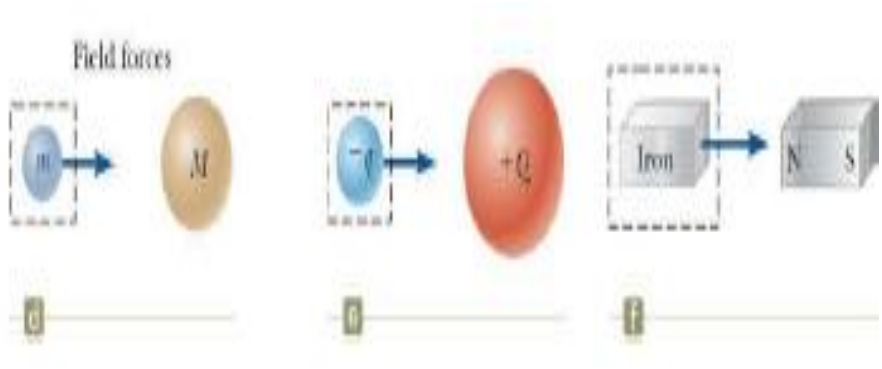
i. Contact forces: involve physical contact between two objects. Examples a, b, c.

e.g. friction, tension, or muscular push.



ii. Field forces: act through empty space. Examples d, e, f.

e.g. Nuclear, Magnetic, Electrical.



Net Force: The vector sum of all forces acting on an object. If the net force is zero,

the object's acceleration is zero (it is either at rest or moving with constant velocity).

- **Fundamental Forces:**
  - ravitational force: Between objects.
  - Electromagnetic forces: Between electric charges.
  - Nuclear force: Between subatomic particles.
  - Weak forces: Arise in certain radioactive decay processes.

Note: These are all field forces

- **Newton's Laws:**.

i. Newton's First Law of Motion (The Law of Inertia):

"In inertial reference frames, objects traveling at constant velocity will maintain that velocity unless acted upon by an outside force; objects at rest will remain at rest unless an outside force acts."

This means that motion does not need a cause, change in motion does. If no net force acts, velocity stays

constant.

Can conclude that any isolated object is either at rest or moving at a constant velocity.

The First Law also allows the definition of force as that which causes a change in the motion of an object.

The tendency of an object to resist any attempt to change its velocity is called inertia.

Inertia: The inherent property of matter to resist changes in its state of motion.

- **Newton's Laws:**. **i. Newton's First Law of Motion** (The Law of Inertia):

Mass is that property of an object that specifies how much resistance an object exhibits to changes in its velocity.

Mass is a scalar quantity. The SI unit of mass is kg.

Mass and weight are two different quantities .

Weight is equal to the magnitude of the gravitational force exerted on the object.

Weight will vary with location.

$m_{earth} = 3Kg,\ m_{moon} = 3\text{Kg} =\approx 1$

$$m_{earth} = 30N,\ w_{moon} \approx 9\text{N}$$

Inertial Frame of Reference: A laboratory (or viewpoint) in which Newton's laws hold true. In an inertial frame, an object's acceleration is always caused by an interaction with another object. Non-inertial frames (e.g., an accelerating car or a spinning Earth) exhibit "fictitious forces" that are not due to physical interactions.

Newton's First Law reveals that force is actually needed to change motion. The natural state of an object is constant velocity (including zero).

- **Biological example:**

A bacterium swimming in water slows down quickly when its flagella stop because viscous drag (a force from the fluid) acts on it.

In vacuum, it would continue moving forever, but inside water, frictional forces dominate.

- **Mass and Gravity:**

Mass: A scalar quantity that measures the amount of matter in an object and its resistance to acceleration (its inertia). It is an intrinsic property and does not change with location. SI Unit: kilogram (kg).

i. Inertial Mass: A measure of an object's resistance to acceleration. It is defined by applying the same force to two different objects and comparing their accelerations:

$$F_1 = F_2 \qquad \frac{m_2}{m_1} = \frac{a_1}{a_2}$$

The more massive object accelerates less.

ii. **Gravitational Mass:** A property of an object that determines the strength of its gravitational interaction.

Rectilinear Motion(Cont.)

Dynamics

- **Mass and Gravity (Cont.):**

- A body's weight (near the Earth) is the gravitational force acting on it or the Earth exerts on that body. Thus, a mass (m) weights
- The weight of an object is the gravitational force :

$$W_{mass\ m} = F_{Earth\ on\ m} = mg_{Earth}$$

($g_{Earth} = 9.8\ m/\ s^2$)

- Mass is an intrinsic property (amount of matter), measured in kg. Weight is a force that depends on location, measured in Newtons (N). A 1 kg mass weighs 9.8 N on Earth but only about 1.6 N on the Moon.
- Biological example:

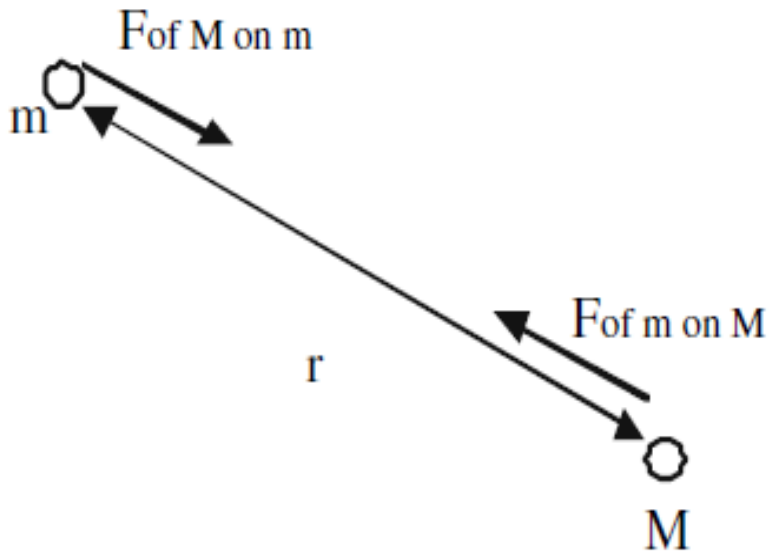
  A 70 kg human has a weight of ≈ 686 N.

- **Mass and Gravity:**

Newton's Law of Universal Gravitation: Every mass attracts every other mass with a force given by:

$$F_{M\ on\ m} = G\frac{Mm}{r^2}$$

where (G) is the gravitational constant, it independent of which masses are interacting and any other physical condition. It is a so called "universal constant" and in SI units its value is close to ($6.67 * 10^{11}\ \mathrm{Nm^2/Kg^2}$), (M and m) are the masses, and (r) is the distance between their centers.



If we divide both sides of Equation by (m) we get the gravitational field produced by M at a distance r from its center: $g_M = G\frac{M}{r^2}$

Free fall: Is the motion of an object under the influence of gravity alone (air resistance is neglected). This is the most important application of constant acceleration.

Acceleration due to gravity (g): For objects near the Earth's surface, g is approximately 9.8 m/s² (≃ 10 m/s²).

Direction: By convention, if we define the upward direction as positive, then the acceleration due to gravity is a (- g = - 9.8 m/s².)

All objects in free fall, regardless of their mass, have the same acceleration (g) when air resistance is negligible.

e.g. If you throw a ball straight up, its velocity decreases by about 9.8 m/s every second until it momentarily stops (v=0) at its highest point. Then, it accelerates downward at the same rate.

ii. Newton's Second Law: The Core Engine of Mechanics

"The net force on a body is equal to the product of the body's mass and its acceleration."

states that the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. Force is the cause of changes in motion, as measured by the acceleration.

Remember, an object can have motion in the absence of forces

$$\Sigma F = ma$$

$\boldsymbol{\Sigma F}$: (Net Force): This is the vector sum of all the individual forces acting on the object, **m**: (Mass): The inertia of the object, **a**: (Acceleration): The resulting acceleration vector.

It's a Vector Equation: This means it holds true for each component independently. $\Sigma F_x = ma_x$, $\Sigma F_y = ma_y$, and $\Sigma F_z = ma_z$.

The net force is the cause the acceleration is the effect. They always point in the same direction.

Newton's Second Law:

If ΣF= 0, then (a = 0): This is Newton's First Law. The Second Law includes the First Law as a special case.

The SI unit of force is Newton, (1 Newton) is the force required to give a (1 kilogram) mass an acceleration of (1 m/s^2), i.e. **1 N = 1 kg·m/s^2**

"In an inertial frame of reference, the acceleration of a body is given by

$$\vec{a} = \frac{\overrightarrow{F_{net}}}{m}$$

That means: **Acceleration is**

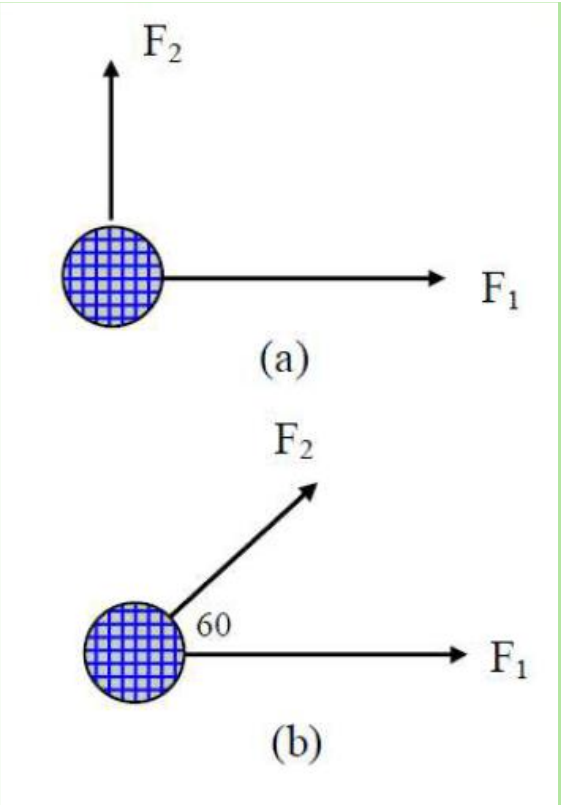
Caused by a net external force. An object cannot accelerate itself.

Directly proportional to the net force (double the force, double the acceleration).

Inversely proportional to mass (double the mass, halve the acceleration for the same force).

Is in the same direction as the net force.

- **Newton's Second Law:**
- In life sciences:
  - The beating heart pushes blood: pressure differences create forces that accelerate blood through arteries.
  - A microorganism's flagellum exerts force to overcome viscous drag, producing forward acceleration.
- **Solution**:



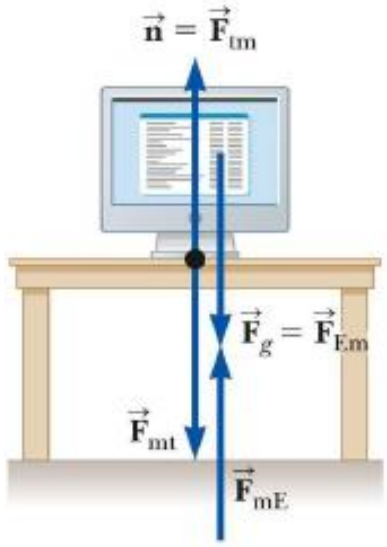
(a) $\sum F = F_1 + F_2 = (\mathbf{20\hat{\imath}} + 15\hat{\jmath})N$ $$\sum \boldsymbol{F} = ma$$ $$\mathbf{20\hat{\imath} + 15\hat{\jmath}} = 5a$$ **a=** $(\mathbf{4\hat{\imath} + 3\hat{\jmath}})\ \boldsymbol{m/s^2} = 5\mathrm{m}/\boldsymbol{s^2}$	**(b)** $\boldsymbol{F}_{2\mathrm{x}} = 15\ cos60 = 7.5\ N$ $\boldsymbol{F_{2y}} = 1560 = 13\ N$ $\boldsymbol{F_2} = (7.5\hat{\imath} + 13\hat{\jmath})\ N$ $\sum \boldsymbol{F} = \boldsymbol{F_1} + \boldsymbol{F_2}$ $= (\mathbf{27.5\hat{\imath} + 13\hat{\jmath}})$ $= ma = 5a$ $\boldsymbol{a} = (\mathbf{5.5\hat{\imath} + 2.6\hat{\jmath}})\boldsymbol{m/s^2} or\ a = 6.08\ m/s^2$

iii. Newton's Third Law of Motion (Action-Reaction)

"If two objects interact When one object exerts a force on a second body, the second exerts a force on the first that is equal in magnitude and opposite in direction".

$$\vec{F}_{2\ on\ 1} = -\vec{F}_{1\ on\ 2}$$

Note on notation :$\vec{F}_{AB}$ {is the force exerted by A on B.



The action force is equal in magnitude to the reaction force and opposite in direction.

One of the forces is the action force, the other is the reaction force.

Example) monitor on table):

The normal force (table on monitor) is the reaction of the force the monitor exerts on the table.

Normal means perpendicular, in this case.

Example) monitor and Earth): The action (Earth on monitor) force is equal in magnitude and opposite in direction to the reaction force, the force the monitor exerts on the Earth.

iii. Newton's Third Law of Motion (Action-Reaction)

- Forces always come in pairs.
- They **do not** cancel each other out because they act on different bodies

e.g:

- Your foot and a soccer ball.
- A runner on the Earth.
- Heart and Blood: When your heart pushes blood forward, the blood pushes backward on the heart walls with equal force, both forces are part of one interaction.

- **Free Body Diagram**

In a free body diagram, you want the forces acting on a particular object.

Model the object as a particle.

The normal force and the force of gravity are the forces that act on the monitor.

The most important step in solving problems involving Newton's Laws is to draw

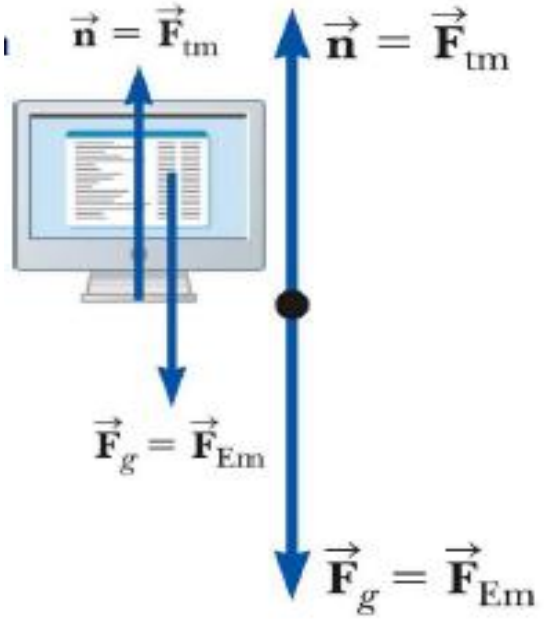
the free body diagram.

Be sure to include only the forces acting on the object of interest.

Include any field forces acting on the object.

Do not assume the normal force equals the weight.

The forces that act on the object are shown as being applied to the dot. The free body helps isolate only those forces acting on the object and eliminate the other forces from the analysis.



- **Diffusion:** Is the random, thermal motion of particles (atoms, molecules, small particles) in a fluid. It is obeying the same physical laws that govern large objects.

- It results from countless random collisions with the fast-moving molecules of the surrounding fluid (e.g., water or air). This is also known as Brownian motion.
- Newtonian Motion: Directed, caused by a net force, displacement is proportional to time

$$\Delta X \; \alpha \; t$$

- Diffusive Motion: Random, no net force, average displacement is zero. However, the mean square displacement grows with time.

Biological Importance: Diffusion is crucial for many life processes, including:

- Gas exchange in the lungs and blood.

Oxygen from the alveoli (high concentration) in to the blood (low concentration), while carbon dioxide diffuses in the

opposite direction

- Nutrient and waste transport across cell membranes (passive transport).

Substance like glucose and amino acid enter the cell, and waste products exit it without needing energy, relying on diffusion

- The motion of non-motile bacteria and molecules within a cell.

Bacteria that lack flagella or cilia, and small molecules within the cell's cytoplasm, move randomly through Diffusion.

- **Applications of Newton's Laws of Motion in One Dimension:**

When acceleration (a) is constant, the definitions of velocity and acceleration can be integrated to yield three key equations that completely describe the motion:

1. Velocity as a function of time:

$$v(\mathrm{t}) = V_0 + at$$

The velocity changes linearly with time.

2. Position as a function of time:

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$$

The position varies quadratically with time

3. Velocity as a function of position:

$V^2 = V_0^2 + 2\mathrm{a}(\mathrm{x} - x_0)$ This equation is useful when time is unknown.

- **Applications of Newton's Laws of Motion in One Dimension:**

When you drop a small object, the constant downward force produces an acceleration .Ignoring air resistance, it falls faster each second.

In life sciences:

When a doctor centrifuges blood, the red blood cells experience a constant centrifugal force outward leading to acceleration until drag balances the force (steady speed).

1.Centrifugal Force: As the machine spins, a strong centrifugal force is generated, pushing the components of the blood outwards (away from the center of rotation). This force is constant as long as the machine is spinning.

2. Acceleration: Initially, the red blood cells (the densest components of

blood) experience this net force, causing them to accelerate outwards. This means their velocity increases.

3. Drag Force and Steady Speed: As the cells accelerate outwards, they encounter resistance or "friction" from the plasma (the liquid part of the blood). This resistance is called the drag force. This drag force increases as the cells' speed increases.

At the point where the centrifugal force (pushing out) balances the drag force (opposing motion), the net force becomes zero.

According to Newton's laws, when the net force is zero, acceleration stops. At this stage, the red blood cells continue to move outward, but at a constant speed (steady speed).